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(54) **VERTICAL SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THE VERTICAL SEMICONDUCTOR DEVICE**

Publication Classification

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(52) **U.S. Cl.**

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(57)

ABSTRACT

A vertical semiconductor device includes: a lower structure; a multi-layer stack structure including a source layer formed over the lower structure and gate electrodes formed over the source layer; a vertical structure penetrating the multi-layer stack structure and including a channel layer insulated from the source layer; a vertical source line spaced apart from the vertical structure to penetrate the multi-layer stack structure and contacting the source layer; and a horizontal source channel contact suitable for coupling the source layer and the channel layer and including a first conductive layer and a second conductive layer that include different dopants.

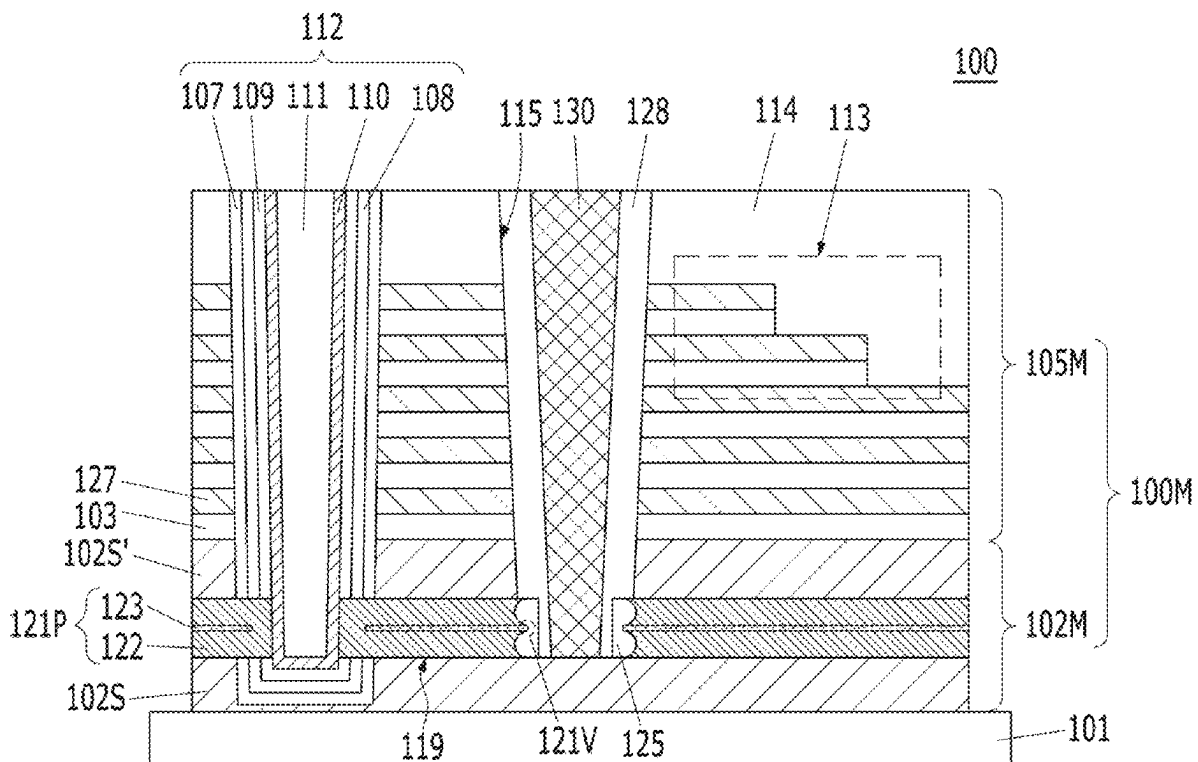


FIG. 1

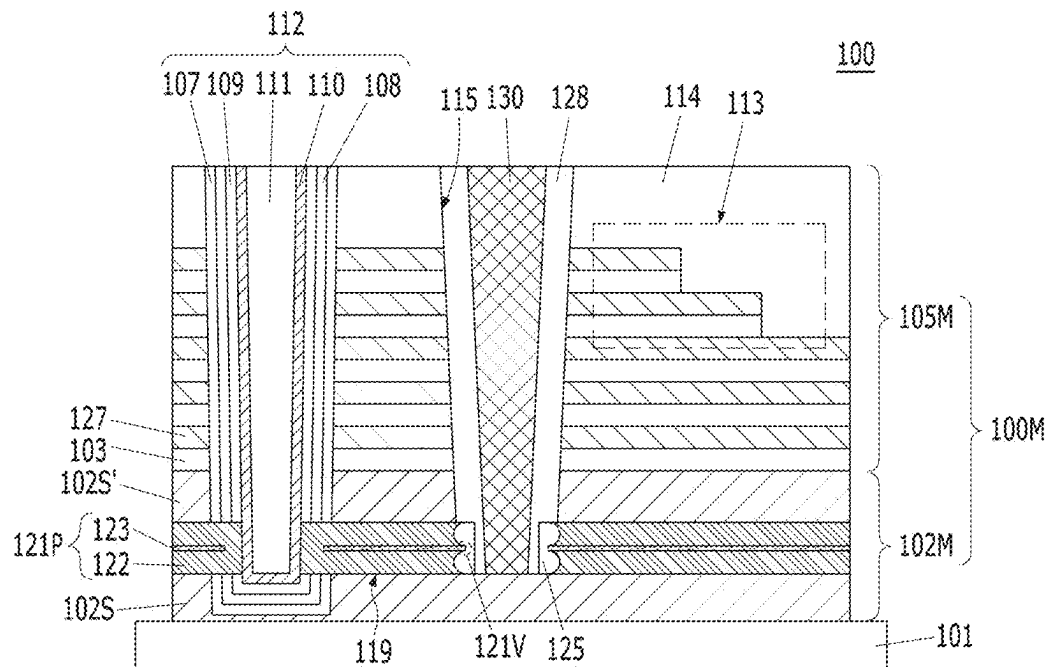


FIG. 2A

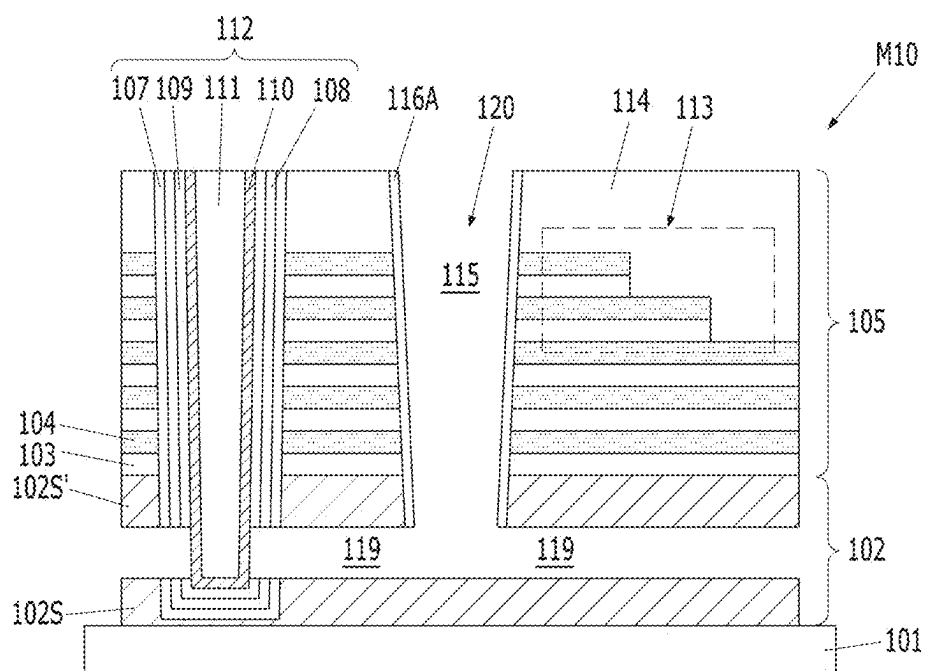


FIG. 2B

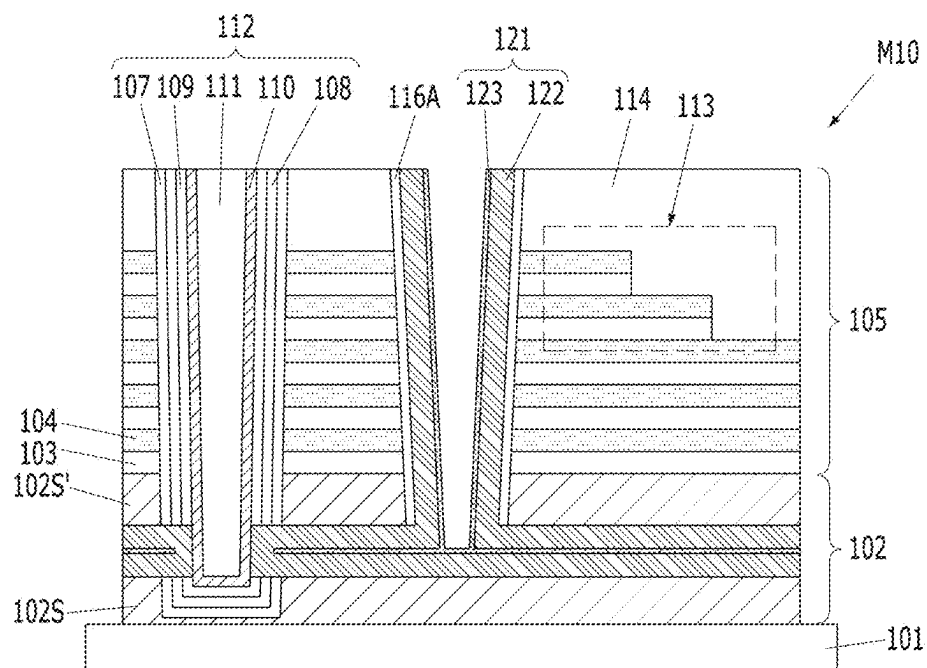
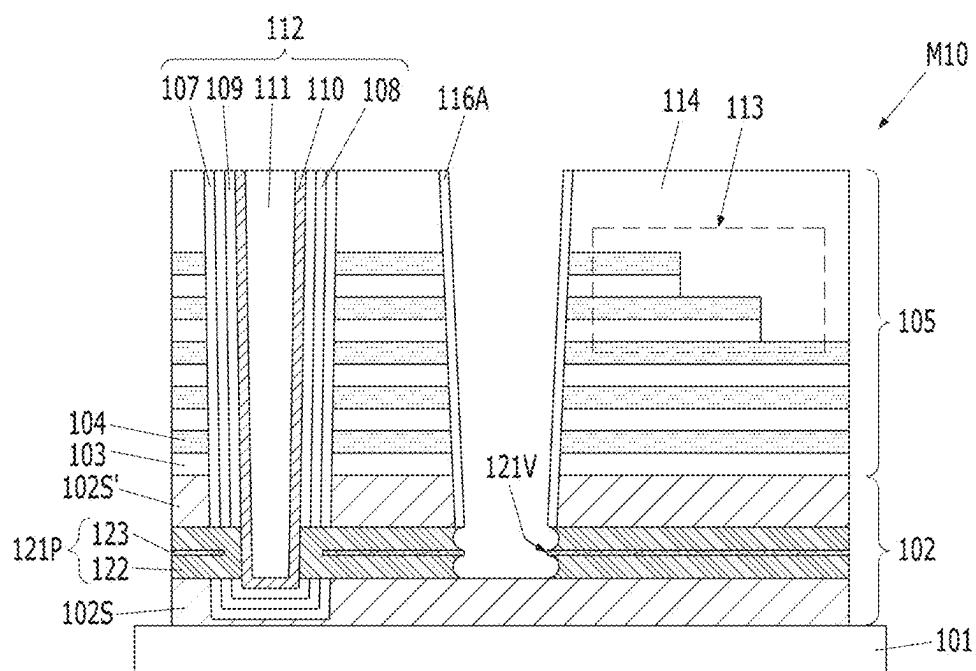


FIG. 2C



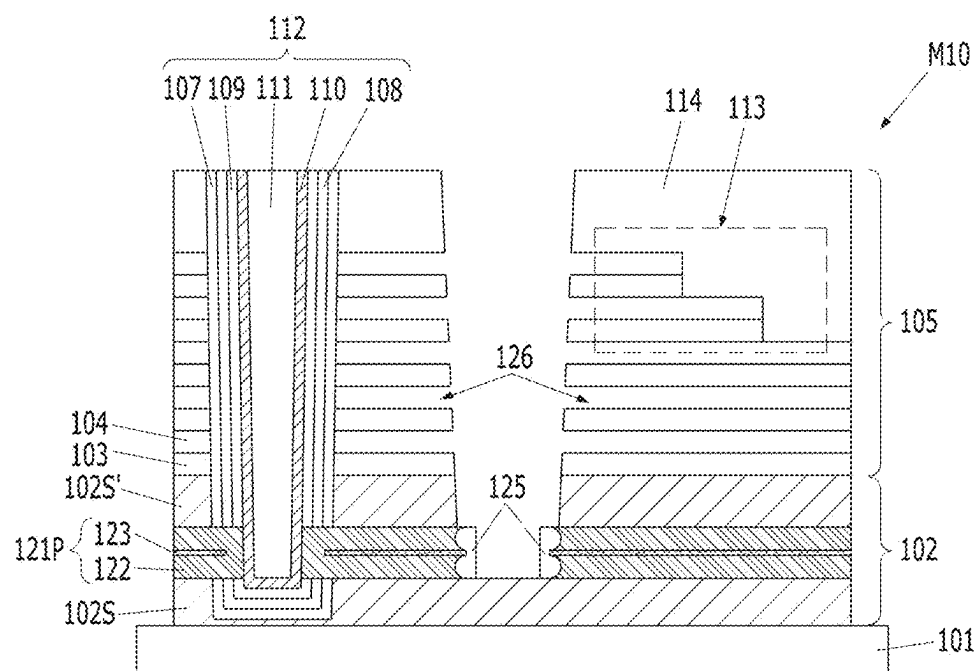


FIG. 2F

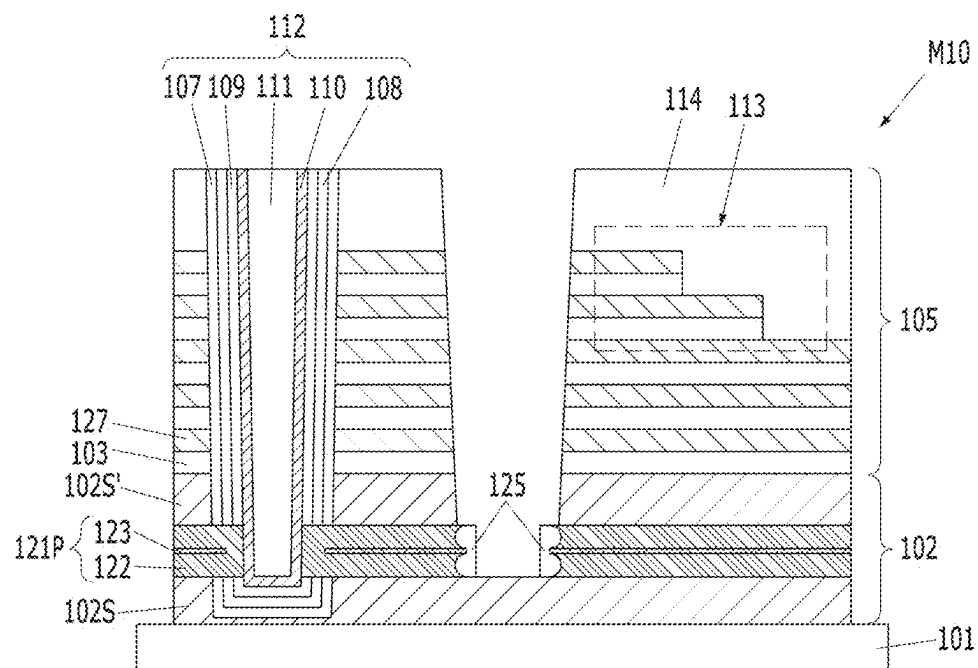


FIG. 2G

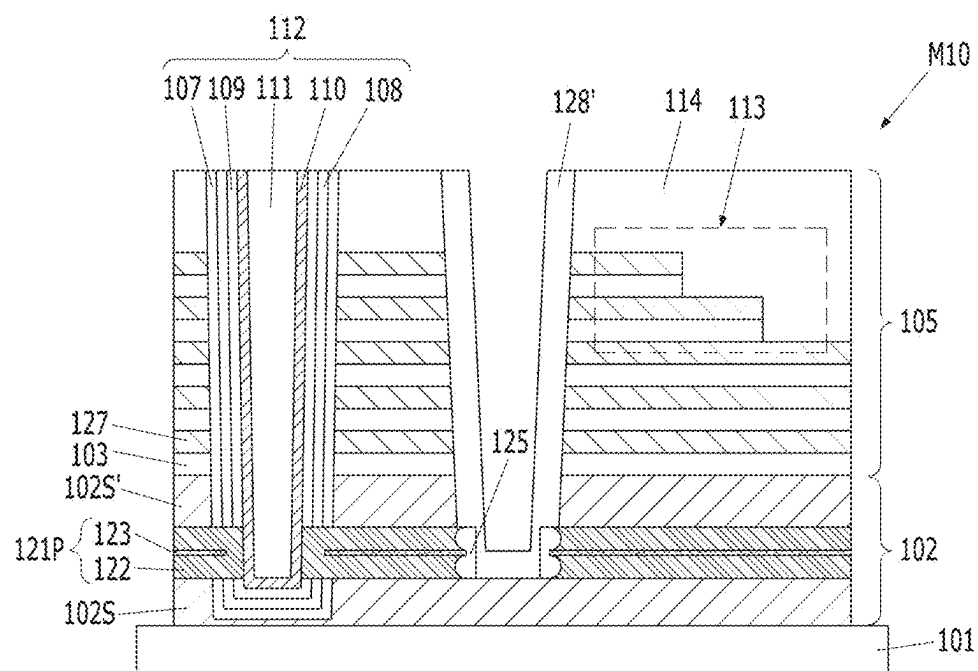


FIG. 2H

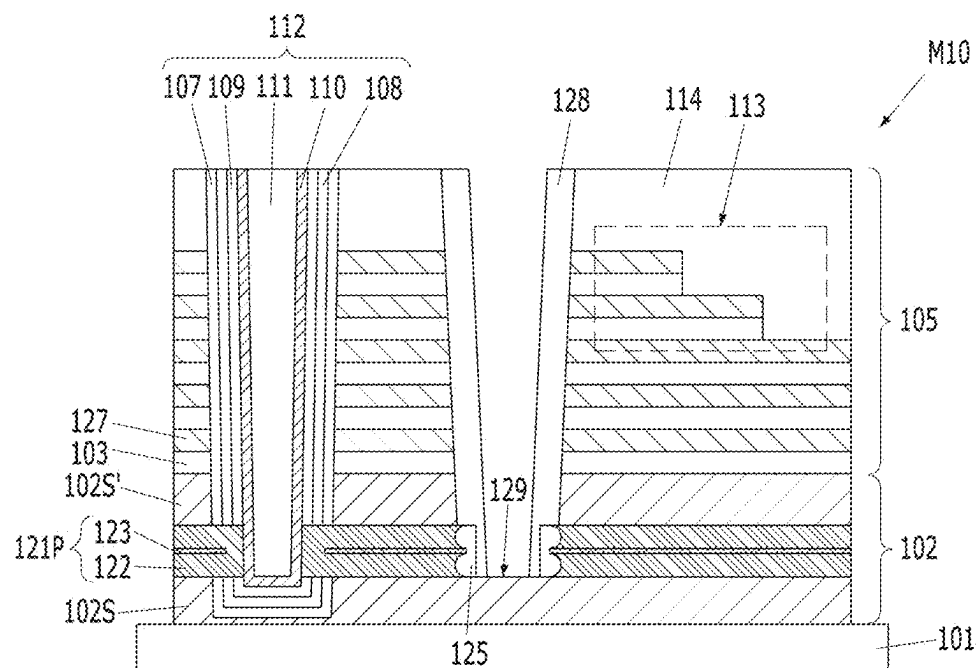


FIG. 2I

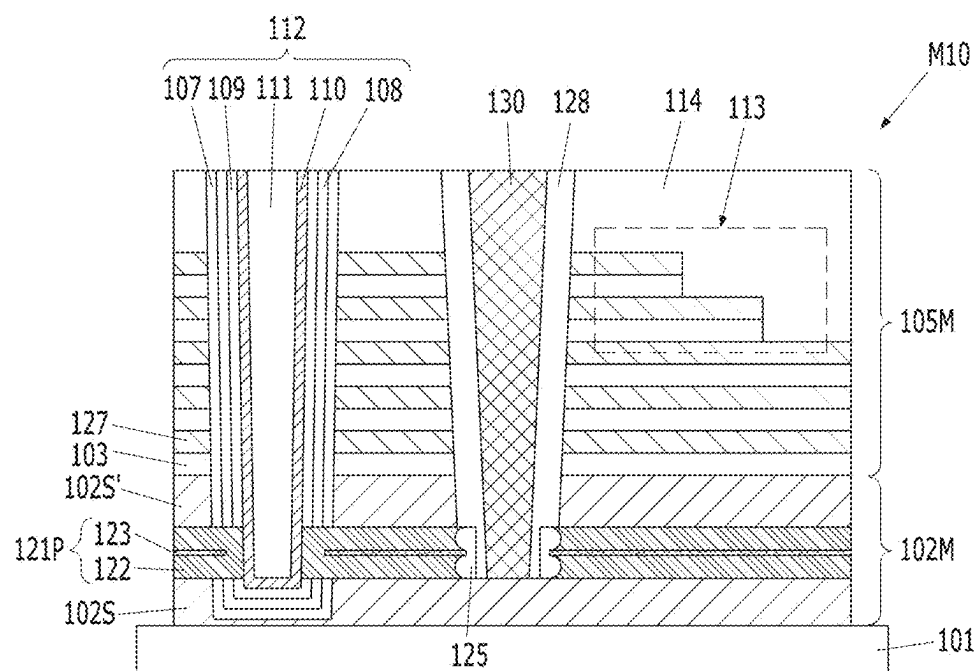


FIG. 3A

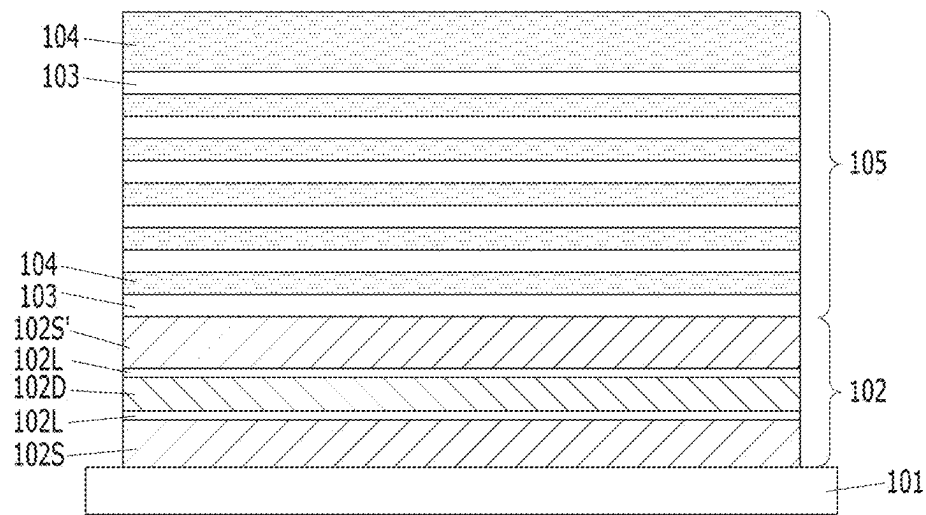


FIG. 3B

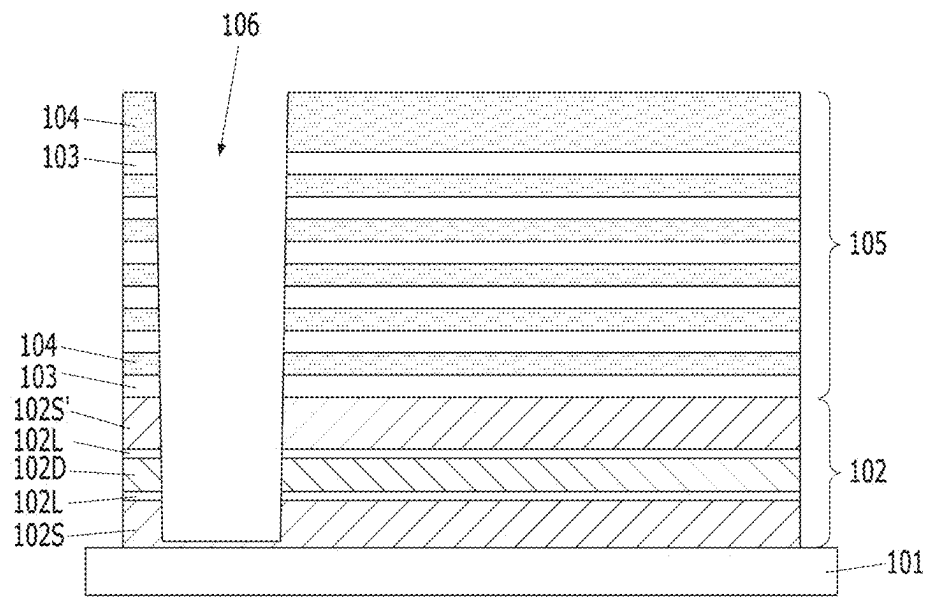


FIG. 3C

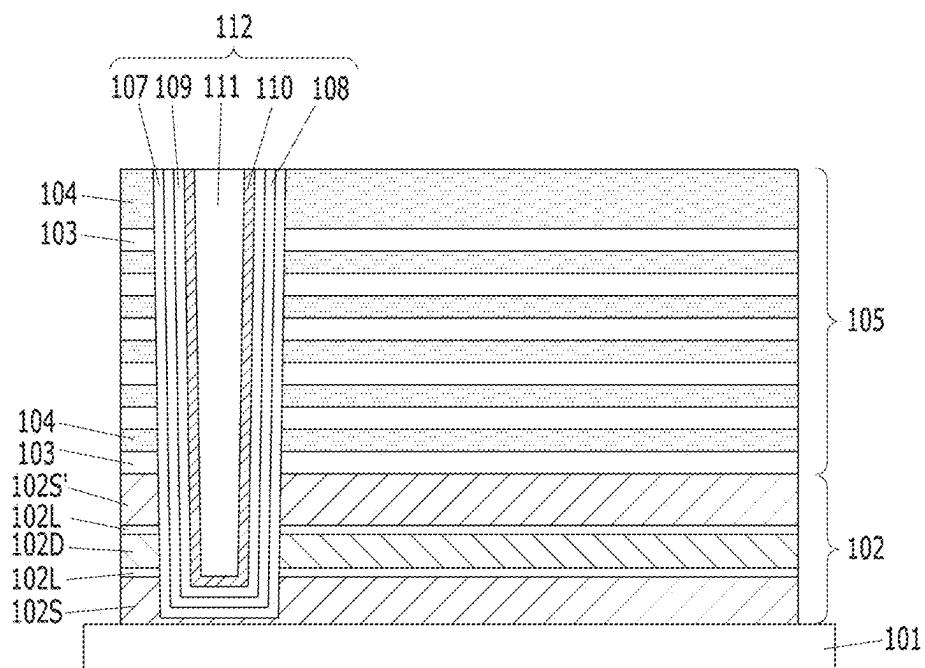


FIG. 3D

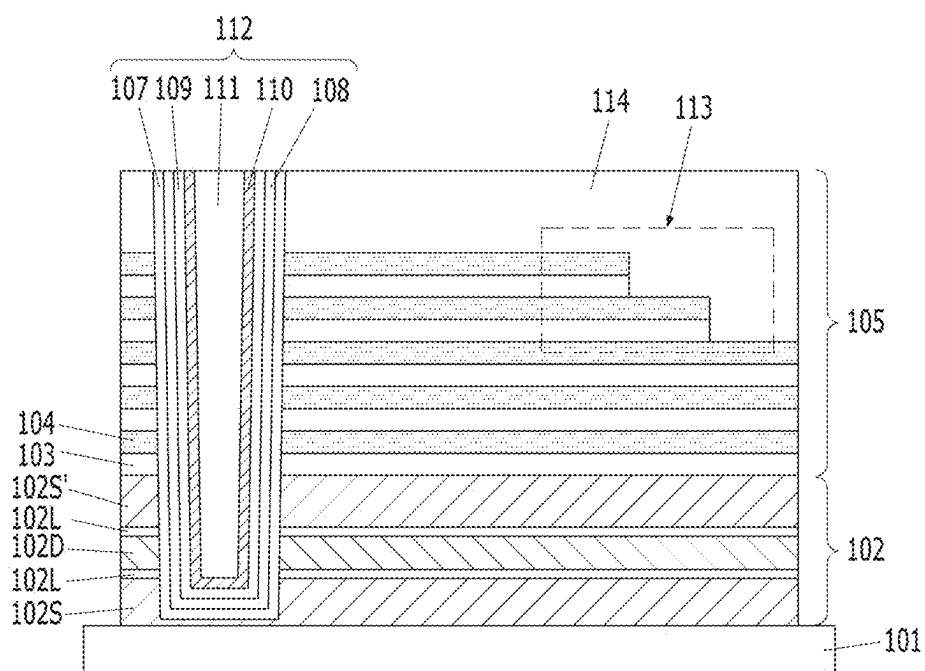


FIG. 3E

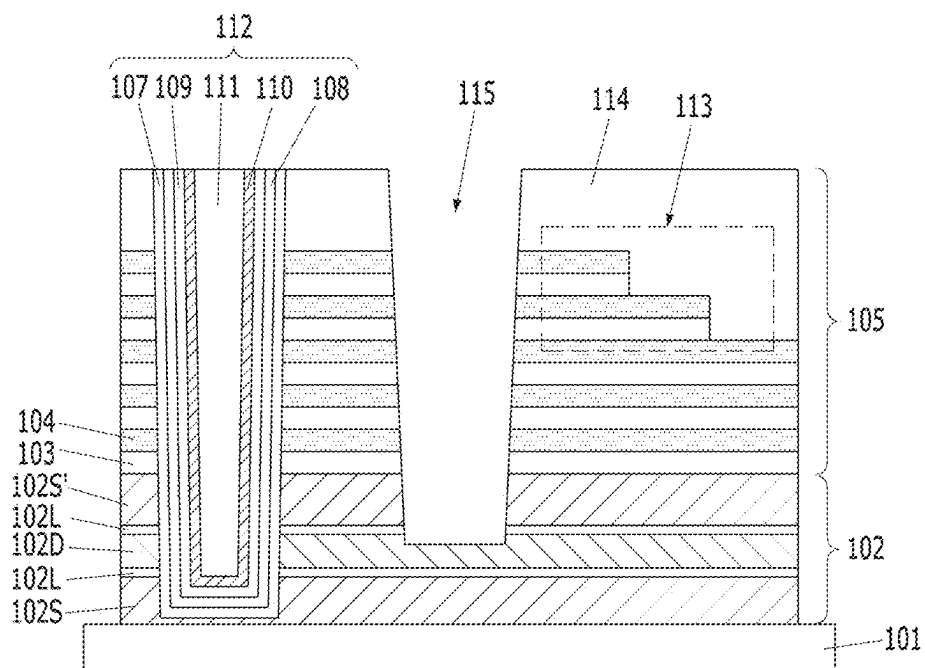


FIG. 3F

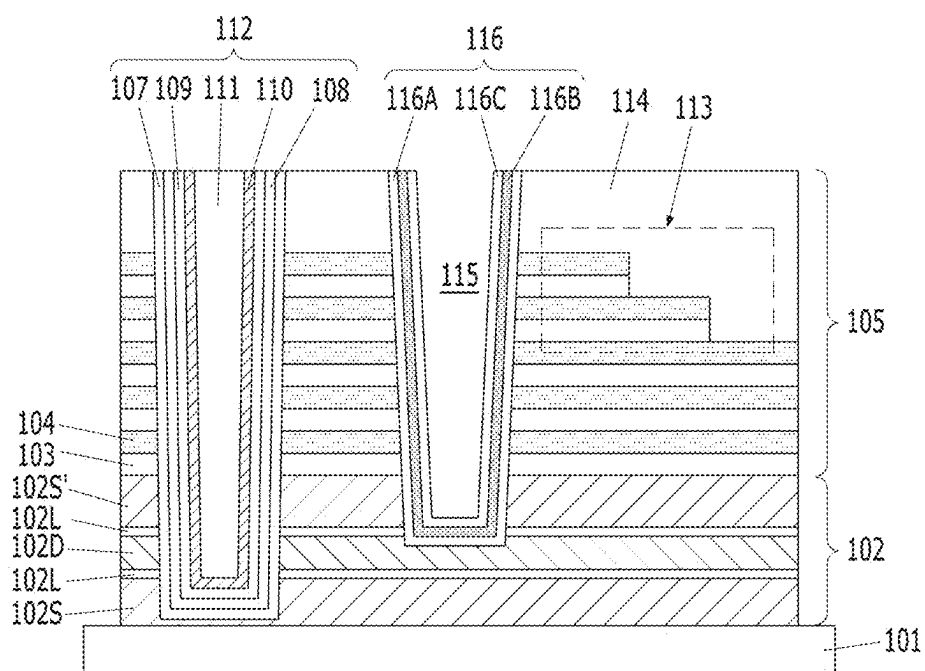


FIG. 3G

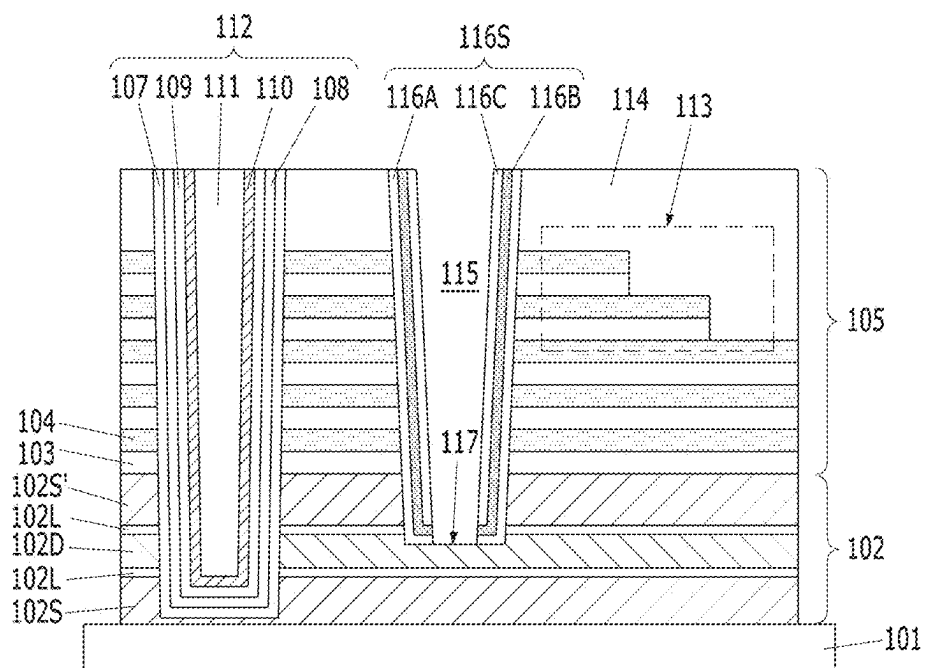


FIG. 3H

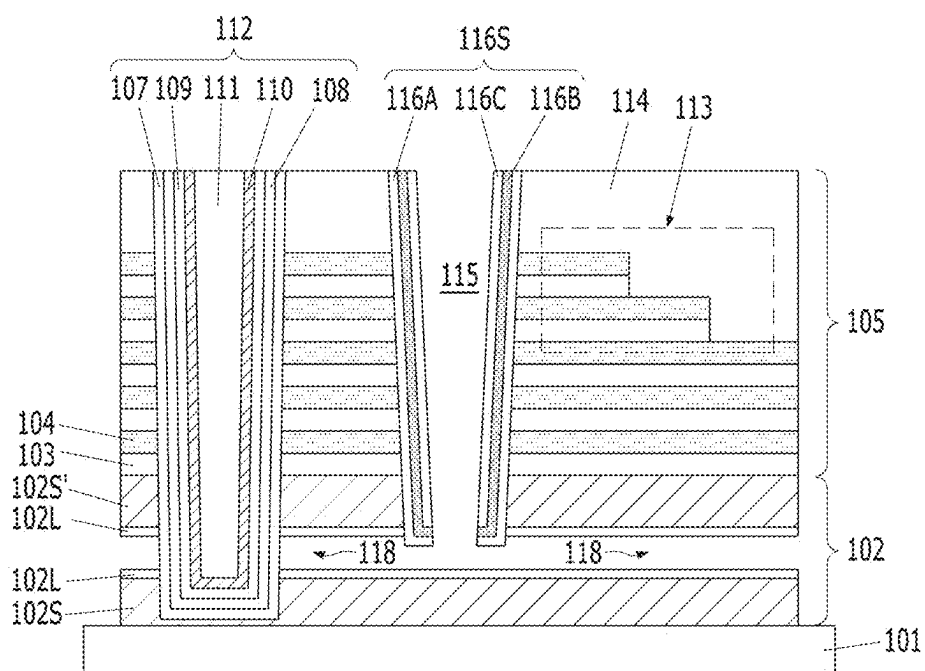


FIG. 3I

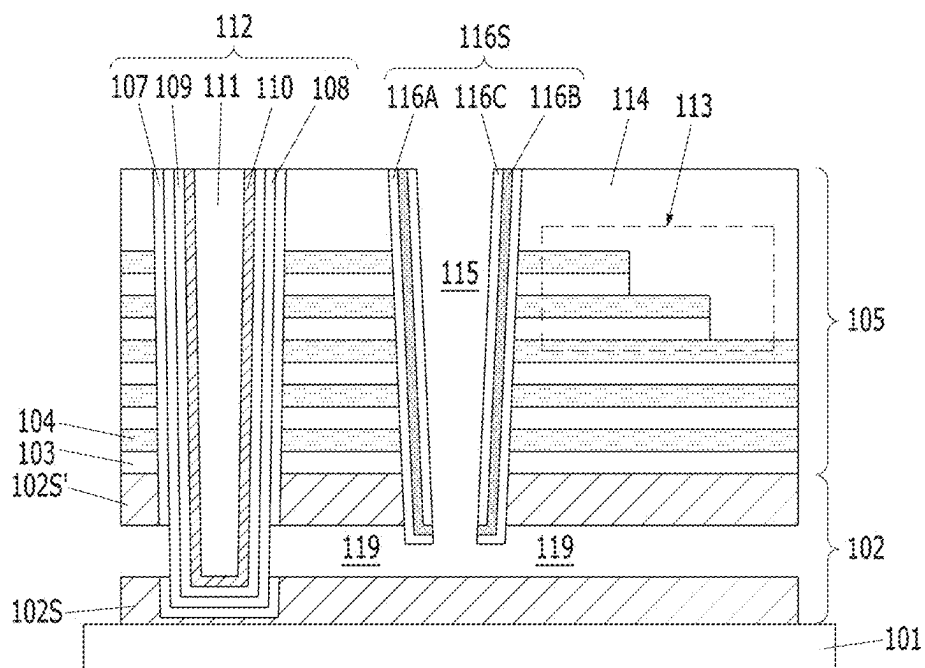


FIG. 3J

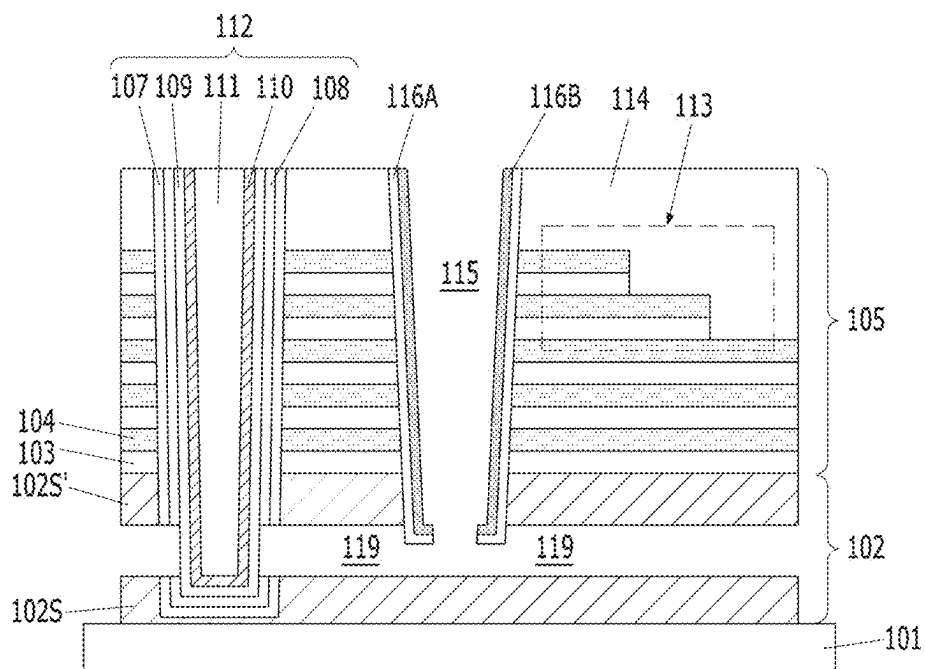


FIG. 4A

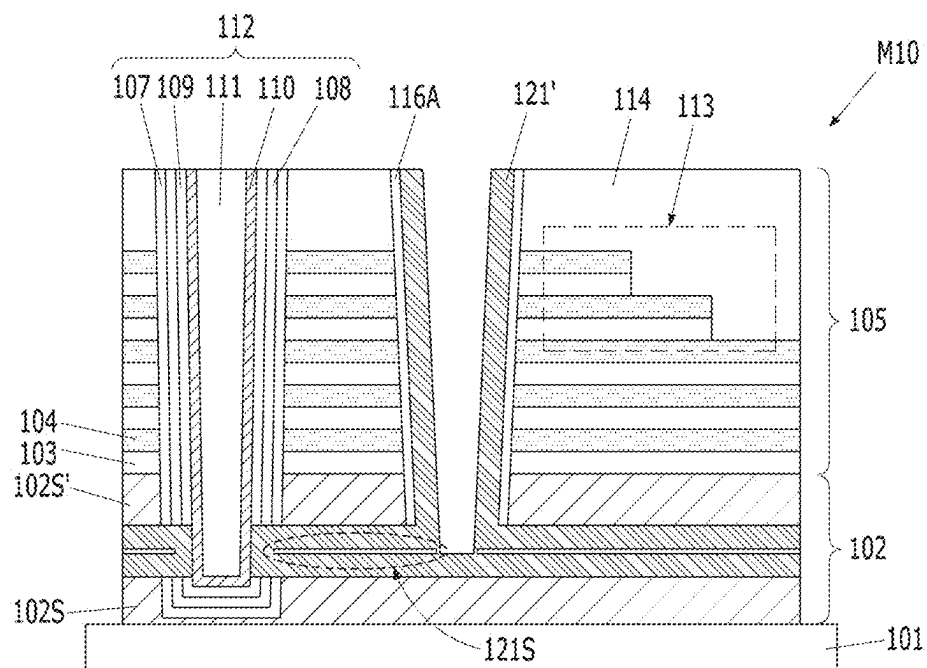


FIG. 4B

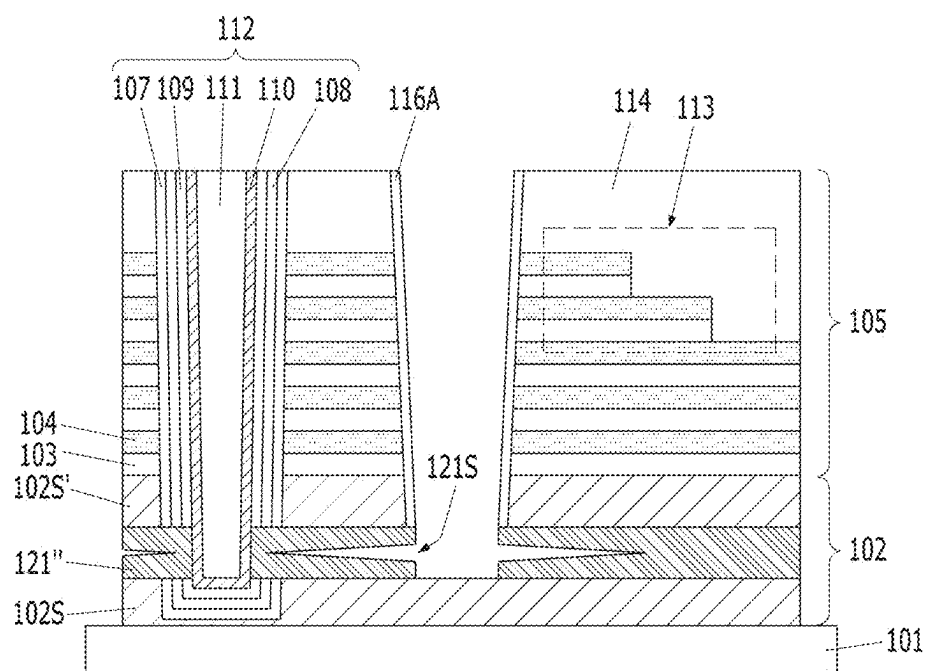


FIG. 4C

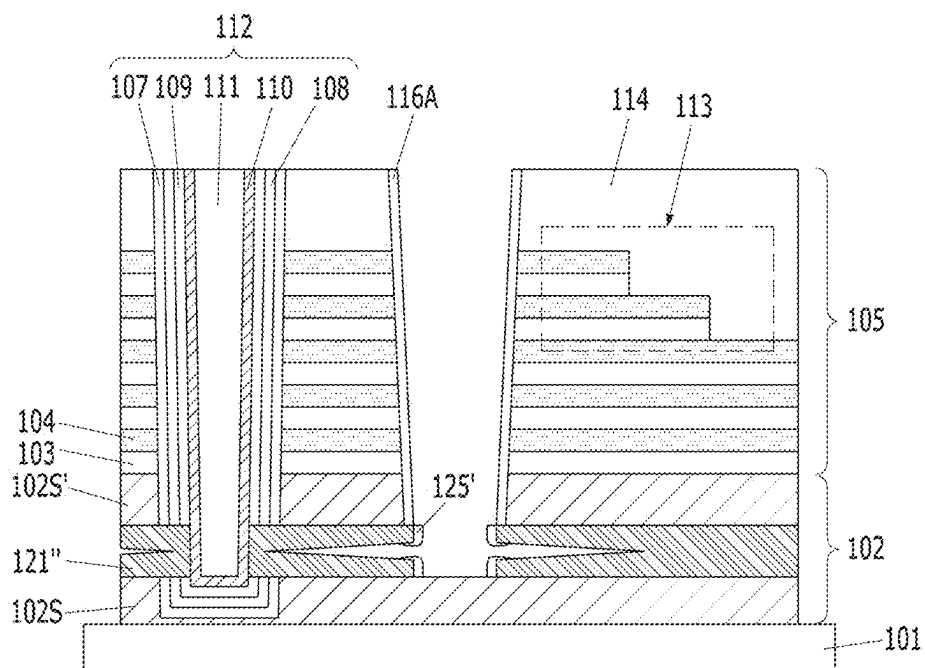
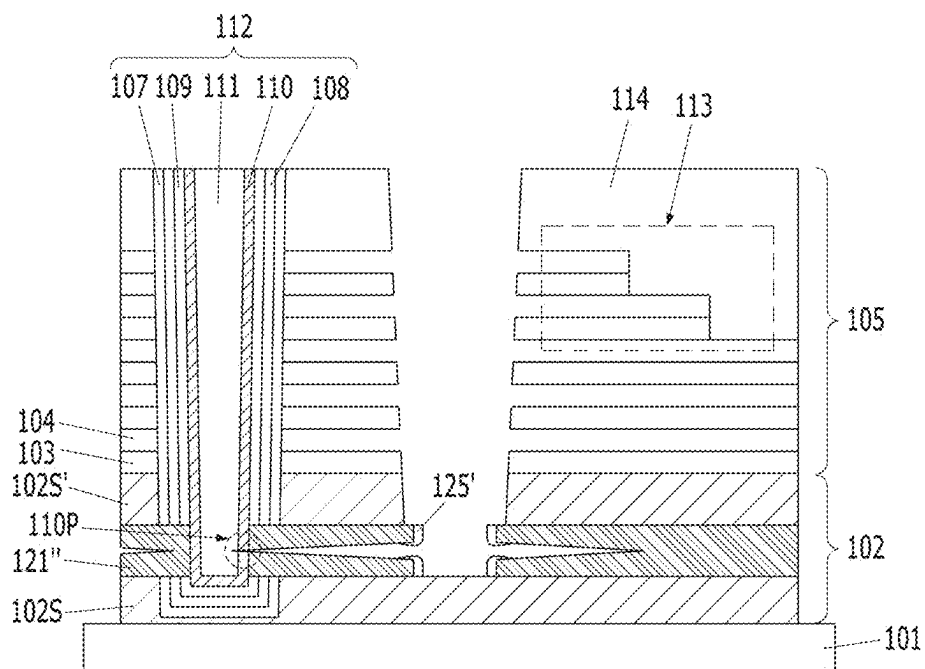


FIG. 4D



VERTICAL SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THE VERTICAL SEMICONDUCTOR DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation application of U.S. patent application Ser. No. 18/446,827, filed on Aug. 9, 2023, which is a continuation application of U.S. patent application Ser. No. 17/567,423, filed on Jan. 3, 2022, which is a continuation application of U.S. patent application Ser. No. 16/680,219, filed on Nov. 11, 2019, which claims priority under 35 U.S.C. § 119(a) to Korean Patent Application No. 10-2019-0042570, filed on Apr. 11, 2019, in the Korean Intellectual Property Office, which applications are incorporated herein by reference in their entirety.

BACKGROUND

1. Technical Field

[0002] Various embodiments generally relate to a semiconductor device, and more particularly, to a vertical semiconductor device and a method for fabricating the vertical semiconductor device.

2. Related Art

[0003] In the manufacturing of electronic devices, such as semiconductor devices, a gap-fill process is required for a three-dimensional structure or a high aspect ratio structure. The gap-filling process of a high aspect ratio structure may be performed, for example, in the fabrication of vertical semiconductor devices.

SUMMARY

[0004] In accordance with an embodiment, a vertical semiconductor device may include: a lower structure; a multi-layer stack structure including a source layer formed over the lower structure and gate electrodes formed over the source layer; a vertical structure penetrating the multi-layer stack structure and including a channel layer insulated from the source layer; a vertical source line spaced apart from the vertical structure to penetrate the multi-layer stack structure and contacting the source layer; and a horizontal source channel contact coupling the source layer and a channel layer and including a first conductive layer and a second conductive layer that include different dopants.

[0005] In accordance with an embodiment, a method for fabricating a vertical semiconductor device may include: forming a first multi-layer stack in which a source sacrificial layer is positioned over a lower source layer, the lower source layer over a lower structure; forming a second multi-layer stack in which dielectric layers and sacrificial layers are alternately stacked over the first multi-layer stack; forming a vertical structure that penetrates the second multi-layer stack and the first multi-layer stack, and includes a channel layer insulated from the lower source layer; forming a vertical contact recess that penetrates the second multi-layer stack and the first multi-layer stack and exposes the source sacrificial layer; forming a horizontal contact recess extending from the vertical contact recess by removing the source sacrificial layer; exposing a side of the channel layer of the vertical structure by extending the horizontal contact recess; and forming a source channel contact that fills the

horizontal contact recess, and includes a first doped layer in contact with the channel layer and a second doped layer doped with a dopant which is different from a dopant of the first doped layer.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a cross-sectional view illustrating a vertical semiconductor device in accordance with an embodiment.

[0007] FIGS. 2A to 2I are cross-sectional views illustrating a method for fabricating a vertical semiconductor device in accordance with an embodiment.

[0008] FIGS. 3A to 3J are cross-sectional views illustrating a method for forming a gap-fill target structure M10.

[0009] FIGS. 4A to 4D are cross-sectional views illustrating a method for fabricating a vertical semiconductor device according to an example.

DETAILED DESCRIPTION

[0010] Various embodiments will be described below with reference to the accompanying drawings. The examples, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete. Throughout the disclosure, like reference numerals refer to like parts throughout the various figures and embodiments.

[0011] The drawings are not necessarily to scale and in some instances, proportions may have been exaggerated in order to clearly illustrate features of the embodiments. When a first layer is referred to as being “on” a second layer or “on” a substrate, it not only refers to a case where the first layer is formed directly on the second layer or the substrate but also a case where a third layer exists between the first layer and the second layer or the substrate.

[0012] In the following embodiments, a high aspect ratio structure may include a three-dimensional structure. The high aspect ratio structure may include a vertical structure, a horizontal structure, or a combination thereof. The high aspect ratio structure may refer to a contact hole, a trench, a recess, or an opening. The high aspect ratio structure may have a height-to-width ratio of at least 5:1 or greater.

[0013] The following embodiments to be described below may be used to gap-fill the horizontal high aspect ratio structure of a vertical semiconductor device, such as a vertical NAND.

[0014] Embodiments may be directed to a gap-filling method that allows a high aspect ratio structure to be gap-filled without voids.

[0015] Embodiments may be directed to a vertical semiconductor device capable of improving reliability and a method for fabricating the vertical semiconductor device.

[0016] FIG. 1 is a cross-sectional view illustrating a vertical semiconductor device in accordance with an embodiment.

[0017] Referring to FIG. 1, the vertical semiconductor device 100 may include a lower structure 101 and a multi-layer stack structure 100M formed over the lower structure 101. The multi-layer stack structure 100M may include a lower stack 102M formed over the lower structure 101 and an upper stack 105M formed over the lower stack 102M.

The multi-layer stack structure **100M** may further include an inter-layer dielectric layer **114** formed over the upper stack **105M**.

[0018] The lower stack **102M** may include source layers **102S** and **102S'** and a source channel contact **121P**. The source layers **102S** and **102S'** may include a lower source layer **102S** and an upper source layer **102S'**. The source channel contact **121P** may be formed between the lower source layer **102S** and the upper source layer **102S'**. The lower source layer **102S** and the upper source layer **102S'** may be of the same material. They may include a semiconductor material such as polysilicon. The source channel contact **121P** may include a semiconductor material such as polysilicon. The source channel contact **121P** may fill the horizontal contact recess **119**. The source channel contact **121P** may be referred to as a horizontal source channel contact **121P**.

[0019] The source channel contact **121P** may include a first conductive layer **122** and a second conductive layer **123**. The first conductive layer **122** and the second conductive layer **123** may have different wet etch rates. The first conductive layer **122** and the second conductive layer **123** may include different dopants. The first conductive layer **122** and the second conductive layer **123** may have a wet etch rate difference by different dopants. The first conductive layer **122** may include a first dopant, and the second conductive layer **123** may include a second dopant. The first dopant may include phosphorus, and the second dopant may include carbon. The first conductive layer **122** may include a phosphorus-doped polysilicon. The second conductive layer **123** may include carbon-doped polysilicon.

[0020] The upper stack **105M** may include an alternating stack in which dielectric layers **103** and gate electrodes **127** are alternately stacked. The dielectric layers **103** may include silicon oxide, and the gate electrodes **127** may include a metal-based material. The gate electrodes **127** may include a stack of tungsten or titanium nitride and tungsten. A step structure **113** may be formed at one end of the gate electrodes **127**. The step structure **113** may include a plurality of steps, and each step may include a pair comprising of a gate electrode **127** and a dielectric layer **103**.

[0021] A vertical structure **112** penetrating the multi-layer stack structure **100M** may be formed. The bottom surface of the vertical structure **112** may extend to contact the lower source layer **102S**. The vertical structure **112** may include a blocking layer **107**, a charge storage layer **108**, a tunnel dielectric layer **109**, and a channel layer **110**. The blocking layer **107** may be in direct contact with the gate electrodes **127**. The vertical structure **112** may further include a core dielectric layer **111** that is formed over the channel layer **110**.

[0022] In the lower portion of the vertical structure **112**, the channel layer **110** may be in direct contact with the source channel contact **121P**. To this end, the blocking layer **107**, the charge storage layer **108**, and the tunnel dielectric layer **109** may be selectively removed from the lower portion of the vertical structure **112**.

[0023] A source line **130** may be formed to be spaced apart from the vertical structure **112** and penetrate the multi-layer stack structure **100M**. A sealing spacer **128** may be formed between the source line **130** and the gate electrodes **127**. The source line **130** may be in direct contact with the lower source layer **102S**. The source channel contact **121P** and the upper source layer **102S'** might not be in contact with the source line **130**. The source line **130** may include a stack of

tungsten or polysilicon and tungsten. The source line **130** may fill the vertical contact recess **115**. The source line **130** may be referred to as a vertical source line. The vertical contact recess **115** and the horizontal contact recess **119** may be coupled to each other. The sealing spacer **128** may be formed on the side wall of the vertical contact recess **115**.

[0024] A barrier oxide **125** may be formed at one end of the source channel contact **121P**. The barrier oxide **125** may be formed between the source channel contact **121P** and a sealing spacer **128**. The barrier oxide **125** may be an oxide that oxidizes the source channel contact **121P**. The barrier oxide **125** may include silicon oxide.

[0025] One end of the source channel contact **121P** may include a concave portion and a convex portion. The convex portion of the source channel contact **121P** may have a peak **121V**. The peak **121V** may be obtained by the wet etch rate difference between the first conductive layer **122** and the second conductive layer **123**. The peak **121V** may be provided by the second conductive layer **123**. In an embodiment, the second conductive layer **123** includes a material having a slower wet etch rate than the first conductive layer **122**.

[0026] Even though the peak **121V** is formed, the surface of the barrier oxide **125** may be uniform. That is, the surface roughness of the barrier oxide **125** opposed to the source line **130** may be planar.

[0027] FIGS. **2A** to **2I** are cross-sectional views illustrating a method for fabricating a vertical semiconductor device in accordance with an embodiment.

[0028] Referring to FIG. **2A**, a gap-fill target structure **M10** may be formed over the lower structure **101**. The gap-fill target structure **M10** may include a first multi-layer stack **102** and a second multi-layer stack **105** over the first multi-layer stack **102**. The gap-fill target structure **M10** may include a source contact opening **120** that penetrates the second multi-layer stack **105** and the first multi-layer stack **102**. The source contact opening **120** may include a vertical contact recess **115** and a horizontal contact recess **119**. The horizontal contact recess **119** may extend from the vertical contact recess **115**. The vertical contact recess **115** may extend in a direction perpendicular to the lower structure **101**, and the horizontal contact recess **119** may extend in a direction horizontal with respect to the lower structure **101**. The gap-fill target structure **M10** may further include the vertical structure **112** penetrating the second multi-layer stack **105** and the first multi-layer stack **102**.

[0029] Hereinafter, a method for forming the gap-fill target structure **M10** will be described with reference to FIGS. **3A** to **3J**.

[0030] Referring to FIG. **3A**, the first multi-layer stack **102** may be formed over the lower structure **101**. The first multi-layer stack **102** may be a stack structure including the lower source layer **102S**, the upper source layer **102S'**, the liner layers **102L**, and a source sacrificial layer **102D**. According to an embodiment, the first multi-layer stack **102** may have a source sacrificial layer **102D** that is formed between the lower source layer **102S** and the upper source layer **102S'** and liner layers **102L** that are formed between the source sacrificial layer **102D** and the lower/upper source layers **102S** and **102S'**. The lower source layer **102S**, the upper source layer **102S'** and the source sacrificial layer **102D** may be of the same material, and the liner layers **102L** may be formed of a material different from those of the lower source layer **102S**, the upper source layer **102S'** and

the source sacrificial layer 102D. The lower and upper source layers 102S and 102S' may have an etch selectivity to the liner layers 102L. The lower source layer 102S, the upper source layer 102S', and the source sacrificial layer 102D may include a semiconductor material, and the liner layers 102L may include a dielectric material. The lower source layer 102S, the upper source layer 102S', and the source sacrificial layer 102D may include polysilicon, and the liner layers 102L may include silicon oxide. The liner layers 102L may be thinner than the lower source layer 102S, the upper source layer 102S', and the source sacrificial layer 102D. For example, the lower source layer 102S and the upper source layer 102S' may have a thickness of approximately 150 nm, and the liner layers 102L may have a thickness of approximately 8 nm. The source sacrificial layer 102D may have the same thickness or may have a thinner thickness than the lower source layer 102S and the upper source layer 102S'.

[0031] Subsequently, a second multi-layer stack 105 may be formed over the first multi-layer stack 102. The second multi-layer stack 105 may be thicker than the first multi-layer stack 102. The second multi-layer stack 105 may include a dielectric layer 103 and a sacrificial layer 104. The second multi-layer stack 105 may include an alternating stack of the dielectric layer 103 and the sacrificial layer 104. The dielectric layer 103 and the sacrificial layer 104 may be alternately stacked alternately several times. The dielectric layer 103 and the sacrificial layer 104 may be of different materials. The dielectric layer 103 may have an etch selectivity to the sacrificial layer 104. The dielectric layer 103 may include silicon oxide, and the sacrificial layer 104 may include silicon nitride. The dielectric layer 103 and the sacrificial layer 104 may be of the same thickness. The dielectric layer 103 and the sacrificial layer 104 may be thicker than the liner layers 102L, and the dielectric layer 103 and the sacrificial layer 104 may be thinner than the lower source layer 102S and the upper source layer 102S'.

[0032] The dielectric layer 103 and the sacrificial layer 104 may be formed by using Chemical Vapor Deposition (CVD) or Atomic Layer Deposition (ALD). The sacrificial layer 104 formed at the top may be attacked and damaged by the subsequent processes. Therefore, the uppermost sacrificial layer 104 may be formed thicker than the other sacrificial layer 104 under the uppermost sacrificial layer 104. The uppermost sacrificial layer 104 may be used as an etch barrier.

[0033] Referring to FIG. 3B, a vertical opening 106 may be formed. The vertical opening 106 may be formed by etching the second multi-layer stack 105 and the first multi-layer stack 102.

[0034] The vertical opening 106 may be formed perpendicular to the surface of the lower structure 101. The vertical opening 106 may be shaped to penetrate the first multi-layer stack 102 and the second multi-layer stack 105. Although not illustrated, from the perspective of a plan view, a plurality of vertical openings 106 may be formed and may have a hole array structure. When the vertical opening 106 is formed, the surface of the lower structure 101 might not be exposed. For example, a portion of the lower source layer 102S may remain below the bottom of the vertical opening 106. According to another embodiment, the vertical opening 106 may be referred to as a 'vertical recess, a vertical hole, or a channel hole'.

[0035] Referring to FIG. 3C, a vertical structure 112 may be formed in the vertical opening 106. The vertical structure 112 may fill the vertical opening 106. The vertical structure 112 may be referred to as a 'pillar structure'.

[0036] The vertical structure 112 may include a blocking layer 107, a charge storage layer 108, a tunnel dielectric layer 109, and a channel layer 110. The vertical structure 112 may include an ONOP structure. The ONOP structure may include a stack of oxide, nitride, oxide, and polysilicon layers. The blocking layer 107 and the tunnel dielectric layer 109 may include an oxide, and the charge storage layer 108 may include a nitride, and the channel layer 110 may include a polysilicon layer. According to an embodiment, the blocking layer 107 may include a high-k material, and the high-k material may include aluminum oxide or hafnium oxide.

[0037] The channel layer 110 may have a cylindrical shape having an inner space. The tunnel dielectric layer 109 may be formed on the outer wall of the channel layer 110, and the charge storage layer 108 may be formed on the outer wall of the tunnel dielectric layer 109. The blocking layer 107 may be formed on the outer wall of the charge storage layer 108.

[0038] The vertical structure 112 may further include a core dielectric layer 111. The inner space of the channel layer 110 may be completely filled with the core dielectric layer 111. The core dielectric layer 111 may include silicon oxide or silicon nitride.

[0039] Referring to FIG. 3D, a step structure 113 including a plurality of steps may be formed. The step structure 113 may be formed by repeatedly performing an etch process of etching the second multi-layer stacks 105 by using a mask and a slimming process of a plurality of masks. The step structure 113 may be formed in a contact region, and the step structure 113 may be formed at one end of the second multi-layer stack 105.

[0040] Subsequently, the uppermost sacrificial layer 104 of the second multi-layer stack 105 may be removed. Then, an inter-layer dielectric layer 114 may be formed over the profile including the step structure 113. Subsequently, the inter-layer dielectric layer 114 may be planarized by a Chemical Mechanical Planarization (CMP) process, and the top surface of the vertical structure 112 may be exposed.

[0041] Although not illustrated, after the inter-layer dielectric layer 114 is formed, a slit may be formed. The slit may be formed in the contact region. The slit may function as a supporter of the contact area.

[0042] Referring to FIG. 3E, an initial vertical contact recess 115 may be formed. The initial vertical contact recess 115 may be formed between the vertical structure 112 and the step structure 113. The initial vertical contact recess 115 may be formed by etching the second multi-layer stack 105, and the initial vertical contact recess 115 may extend downward to a portion of the first multi-layer stack 102. The bottom surface of the initial vertical contact recess 115 may penetrate the upper source layer 102S' and the uppermost liner layer 102L. The etch process for forming the initial vertical contact recess 115 may stop above the source sacrificial layer 102D. The upper surface of the source sacrificial layer 102D may be partially recessed while the initial vertical contact recess 115 is formed. The initial vertical contact recess 115 may be referred to as a slit or trench. From the perspective of a top view, the initial vertical contact recess 115 may have a shape of a line extending in one direction.

[0043] The initial vertical contact recess 115 may have a high aspect ratio perpendicular to the surface of the lower structure 11.

[0044] Referring to FIG. 3F, a sacrificial sealing layer 116 may be formed over the initial vertical contact recess 115. The sacrificial sealing layer 116 may include a first sealing layer 116A, a second sealing layer 116B, and a third sealing layer 116C. The first sealing layer 116A and the third sealing layer 116C may be of the same material, and the second sealing layer 116B may be of a material different from the materials of the first and third sealing layers 116A and 116C. The second sealing layer 116B may have an etch selectivity to the first and third sealing layers 116A and 116C. The first sealing layer 116A and the third sealing layer 116C may include an oxide, and the second sealing layer 116B may include a nitride. Thus, the sacrificial sealing layer 116 may include a NON structure. The NON structure may refer to a stack of a nitride, an oxide, and a nitride. The bottom and side walls of the initial vertical contact recess 115 may be covered with the sacrificial sealing layer 116.

[0045] Referring to FIG. 3G, a cutting process 117 of the sacrificial sealing layer 116 may be performed. The cutting process 117 of the sacrificial sealing layer 116 may include an etch process, and the surface of the source sacrificial layer 102D may be exposed by the cutting process 117. After the cutting process 117 is performed, the sacrificial sealing layer 116 may remain as sacrificial spacers 116S on both side walls of the initial vertical contact recess 115. The cutting process 117 of the sacrificial sealing layer 116 may include an etch-back process.

[0046] Referring to FIG. 3H, the source sacrificial layer 102D may be selectively removed through the initial vertical contact recess 115. As a result, an initial horizontal contact recess 118 may be formed. The initial vertical contact recess 115 and the initial horizontal contact recess 118 may be coupled to each other. The initial horizontal contact recesses 118 may be formed between the liner layers 102L by removing the source sacrificial layer 102D by a dip-out process. The initial horizontal contact 118 may be parallel to the surface of the lower structure 101. When the source sacrificial layer 102D is removed, the liner layers 102L having etch selectivity may remain unremoved. The initial horizontal contact recesses 118 may be formed in the first multi-layer stack 102. When the source sacrificial layer 102D is removed, the lower source layer 102S and the upper source layer 102S' might not be removed. A wet etch process may be applied to remove the source sacrificial layer 102D. Since the source sacrificial layer 102D includes a polysilicon layer, the wet etch process may be performed by using a chemical capable of etching the polysilicon layer.

[0047] A portion of the initial horizontal contact recess 118 may expose the side walls of the lower portion of the vertical structure 112. The outer wall of the vertical structure 112 may be the blocking layer 107, and the blocking layer 107 of the vertical structure 112 may be exposed by the initial horizontal contact recess 118. Since the blocking layer 107 contains an oxide, the blocking layer 107 of the vertical structure 112 having an etch selectivity might not be etched while the source sacrificial layer 102D is removed. From the perspective of a top view, the initial horizontal contact recess 118 may have a shape of surrounding the side walls of the lower portion of the vertical structure 112.

[0048] Referring to FIG. 3I, the liner layers 102L may be removed. Thus, the height of the initial horizontal contact

recess 118 may be increased. The height-widened initial horizontal contact recess may be formed with a reference numeral 119. Hereinafter, the height-widened initial horizontal contact recess may be referred to as a horizontal contact recess 119.

[0049] While the liner layers 102L are removed, the blocking layer 107 of the vertical structure 112 may be removed. As a result, the charge storage layer 108 of the vertical structure 112 may be exposed. A wet etch process may be applied to remove the liner layers 102L. The wet etch process may be performed by using a chemical capable of selectively removing a silicon oxide.

[0050] Referring to FIG. 3J, the charge storage layer 108 of the vertical structure 112 may be removed through the initial vertical contact recess 115 and the horizontal contact recess 119. When the charge storage layer 108 is removed, the third sealing layer 116C of the sacrificial spacer 116S may be removed. The charge storage layer 108 and the third sealing layer 116C may be removed by a wet etch process. Since the charge storage layer 108 and the third sealing layer 116C include a nitride, the wet etch process may use a chemical capable of etching the nitride.

[0051] The horizontal-direction length of the horizontal contact recess 119 may be increased by removing the charge storage layer 108. The horizontal-direction width of the initial vertical contact recess 115 may be increased by removing the third sealing layer 116C. Hereinafter, the widened initial vertical contact recess 115 may be referred to as a vertical contact recess 115.

[0052] The tunnel dielectric layer 109 of the vertical structure 112 may then be removed through the vertical contact recess 115 and the horizontal contact recess 119. While the tunnel dielectric layer 109 is removed, the second sealing layer 116B of the sacrificial spacer 116S may be removed. The tunnel dielectric layer 109 and the second sealing layer 116B may be removed by a wet etch process. Since the tunnel dielectric layer 109 and the second sealing layer 116B include an oxide, the wet etch process may use a chemical capable of etching the oxide.

[0053] The horizontal-direction length of the horizontal contact recess 119 may be increased by removing the tunnel dielectric layer 109. The horizontal-direction width of the vertical contact recess 115 may be increased by removing the second sealing layer 116B.

[0054] FIG. 2A shows a result after the tunnel dielectric layer 109 is removed. The first sealing layer 116A may remain on the side walls of the vertical contact recess 115. The horizontal contact recess 119 may extend from the vertical contact recess 115.

[0055] The initial vertical contact recess 115 and the initial horizontal contact recess 118 may be respectively widened by a series of processes as described above.

[0056] As described above, the source contact openings 120 penetrating the first multi-layer stack 102 and the second multi-layer stack 105 may be formed by the series of the processes shown in FIGS. 3A to 3J. The source contact openings 120 may include the vertical contact recess 115 and the horizontal contact recess 119. The horizontal contact recess 119 may extend from the vertical contact recess 115. The vertical contact recess 115 may extend in a direction perpendicular to the lower structure 101, and the horizontal contact recess 119 may extend in a horizontal direction with respect to the lower structure 101. The horizontal contact recess 119 may have a high aspect ratio that is parallel to the

surface of the lower structure 101. The vertical contact recess 115 may have a high aspect ratio perpendicular to the lower structure 101.

[0057] Referring to FIG. 2B, a conductive layer 121 may be formed over the gap-fill target structure M10 including the source contact openings 120. The conductive layer 121 may include a first conductive layer 122 and a second conductive layer 123 that are sequentially stacked. The first conductive layer 122 may include a first dopant, and the second conductive layer 123 may include a second dopant. The first dopant and the second dopant may be different materials. For example, the first conductive layer 122 may include phosphorus (P), and the second conductive layer 123 may include carbon. The first conductive layer 122 and the second conductive layer 123 may be deposited in situ. The first conductive layer 122 may include phosphorus-doped polysilicon (SiP), and the second conductive layer 123 may include a carbon-doped polysilicon (SiC).

[0058] The wet etch rate of carbon-doped polysilicon (SiC) may be lower than the wet etch rate of the phosphorus-doped polysilicon (SiP). For example, the wet etch rate of the carbon-doped polysilicon (SiC) may be $\frac{1}{5}$ slower than the wet etch rate of the phosphorus-doped polysilicon (SiP). The difference in the wet etch rate may improve the etch profile of the conductive layer 121. For example, when the conductive layer 121 is formed only of the phosphorus-doped polysilicon (SiP), lateral cutting of a V shape may occur after the conductive layer 121 is etched. On the other hand, when the conductive layer 121 is formed of two layers of the carbon-doped polysilicon (SiC) and the phosphorus-doped polysilicon (SiP), the lateral cutting after the etching of the conductive layer 121 may be prevented. Also, when the conductive layer 121 is formed of two layers of the carbon-doped polysilicon (SiC) and the phosphorus-doped polysilicon (SiP), a protruding shape may be formed after the conductive layer 121 is etched. The protruding shape may prevent less oxidation from the very source.

[0059] Referring to FIG. 2C, the conductive layer 121 may be selectively removed. The selective removal process of the conductive layer 121 may include a recessing process, and the recessing process may include wet etching. The source channel contact 121P may be formed in the horizontal contact recess 119 by the wet etching of the conductive layer 121. The source channel contact 121P may include a stack of a first conductive layer 122 and a second conductive layer 123. The etched surface of the source channel contact 121P may include a recess and a peak 121V without lateral cutting. The peak 121V may be provided by the second conductive layer 123, and the recess may be provided by the first conductive layer 122. When wet etching is performed on the conductive layer 121, the etch rate of the second conductive layer 123 may be slower than that of the first conductive layer 122. Therefore, the peak 121V may be formed.

[0060] Referring to FIG. 2D, the source channel contact 121P may be exposed to an oxidation process 124. The oxidation process 124 may include dry oxidation or wet oxidation. The etched surfaces of the source channel contacts 121P may be oxidized by the oxidation process 124. Thus, a barrier oxide 125 may be formed at the end of the source channel contact 121P. The barrier oxide 125 may include silicon oxide.

[0061] The barrier oxide 125 may be formed with a stable thickness. Herein, the stable thickness may refer to a con-

tinuous thickness. For example, the stable thickness of the barrier oxide 125 may be a thickness that may protect the source channel contact 121P from being attacked during a subsequent process.

[0062] Meanwhile, when the source channel contact 121P is formed of only phosphorus-doped polysilicon (SiP), the barrier oxide 125 may be formed in a discontinuous thickness. In other words, even though the barrier oxide 125 is formed, the etched surface of the source channel contact 121P may be partially exposed.

[0063] Referring to FIG. 2E, the sacrificial layer 104 of the second multi-layer stack 105 may be selectively removed. Accordingly, the horizontal gate recesses 126 between the dielectric layers 103 may be formed. Since the sacrificial layer 104 includes a nitride, the sacrificial layer 104 may be removed by a chemical containing phosphoric acid (H_3PO_4). When the sacrificial layer 104 is removed, the remaining first sealing layer 116A may also be removed at the same time.

[0064] During the removal of the sacrificial layer 104, the source channel contact 121P may be fully protected by the barrier oxide 125. The barrier oxide 125 may serve as a protective layer to protect the source channel contact 121P from being attacked.

[0065] Meanwhile, when the barrier oxide 125 is formed in a discontinuous thickness, the chemical may damage the source channel contact 121P. When the source channel contact 121P is damaged, there may be a problem in that the channel layer 110 of the vertical structure 112 may be broken.

[0066] Referring to FIG. 2F, the gate electrodes 127 may fill the horizontal gate recesses 126. The gate electrodes 127 may be formed by depositing a conductive material to fill the horizontal gate recesses 126 and then performing an etch-back process. The gate electrodes 127 may include a low resistance material. The gate electrodes 127 may be a metal-based material. The gate electrodes 127 may include a metal, a metal silicide, a metal nitride, or a combination thereof. For example, the metal may include nickel, cobalt, platinum, titanium, tantalum or tungsten. The metal silicide may include nickel silicide, cobalt silicide, platinum silicide, titanium silicide, tantalum silicide or tungsten silicide. The gate electrodes 127 may include a stack of titanium nitride and tungsten.

[0067] Referring to FIG. 2G, a sealing layer 128' may be formed. One end of one side of the gate electrodes 127 may be sealed. The sealing layer 128' may cover the barrier oxide 125. The sealing layer 128' may include silicon oxide or silicon nitride.

[0068] Referring to FIG. 2H, the sealing layer 128' may be selectively removed so that the surface of the lower source layer 102S is exposed (reference numeral 129'). Thus, a sealing spacer 128 may be formed on the side walls of the vertical contact recess 115. The bottom portion of the ceiling spacer 128 may completely cover the side walls of the barrier oxide 125.

[0069] Referring to FIG. 2I, a source line 130 may be formed. The source line 130 may fill the vertical contact recess 115. The source line 130 may be a stack of polysilicon and tungsten. According to an embodiment, the source line 130 may be formed of tungsten.

[0070] As described above, the lower stack 102M and the upper stack 105M may be formed over the lower structure 101. The lower stack 102M may include source layers 102S

and source channel contacts **121P**. The upper stack **105M** may be formed by alternately stacking the dielectric layers **103** and the gate electrodes **127**. The vertical structure **112** may penetrate the upper stack **105M** and the lower stack **102M**. The lower portion of the channel layer **110** of the vertical structure **112** may directly contact the source channel contact **121P**. The channel layer **110**, the lower source layer **102S**, and the source line **130** may be electrically coupled to each other through the source channel contact **121P**.

[0071] According to the above description, in an embodiment, since the source channel contact **121P** is formed by using the first conductive layer **122** and the second conductive layer **123** having different wet etch rates, it may be possible to remove a weak point that may cause less oxidation. Thus, the barrier oxide **125** may be formed with a stable thickness, and may protect the source channel contact **121P** from being attacked by the subsequent process.

[0072] In addition, even though the source channel contact **121P** is attacked not at its seam but at other points, the seam of the source channel contact **121P** coupled to the channel layer **110** may be reinforced. Therefore, it may be possible to prevent defects that may occur due to discontinuous channel layer **110**.

[0073] According to the embodiments described above, the carbon concentration of the second conductive layer **123** may be reduced to adjust the size and shape of the peak **121V**.

[0074] FIGS. 4A to 4D are views for explaining a method of fabricating a vertical semiconductor device according to an example. The method of fabricating a vertical semiconductor device according to the example may be similar to the processes shown in FIGS. 2A to 2I. However, in the example, polysilicon having the same dopant may be deposited without applying polysilicon having a different dopant when the conductive layer **121** is deposited.

[0075] Referring to FIG. 4A, a conductive layer **121'** may be formed over the gap-fill target structure **M10** including a vertical recess and a horizontal recess. The conductive layer **121'** may be deposited with polysilicon having the same dopant. In the example, a seam **121S'** may exist on the interface of polysilicon during deposition of the conductive layer **121'**. The polysilicon with the same dopant may have the same wet etch rate. The conductive layer **121'** may be formed of phosphorus-doped polysilicon.

[0076] Subsequently, referring to FIG. 4B, the conductive layer **121'** may be etched by a wet etch process to form a source channel contact **121''**. According to the example, an undercut **121S** may be formed by the seam **121S'** on the etched surface of the source channel contact **121''**. Such an undercut **121S** may be generated by etching the conductive layer **121'** having the same wet etch rate.

[0077] Referring to FIG. 4C, a barrier oxide **125'** may be formed on the etched surface of the source channel contact **121''**. According to the example, the barrier oxide **125'** may be formed in a discontinuous thickness. Thus, the source channel contact **121''** may be attacked in the subsequent process because the barrier oxide **125'** does not protect the undercut **121S**.

[0078] Referring to FIG. 4D, the sacrificial layers **103** may be removed. While the sacrificial layers **103** are removed, the source channel contact **121''** may be damaged by the undercut **121S**. When the source channel contact **121''** is damaged, the channel layer **110** may be discontinuous **110P**.

[0079] As described above, in the example, in the process of depositing the conductive layer **121'**, there may be interface between polysilicon, and the etch rate may be accelerated due to the presence of the seam **121S'** that is formed on the interface of the polysilicon when the subsequent wet etch process is performed. As the wet etch process is performed rapidly, the undercut **121S** may be formed in which the etch profile at the position of the seam penetrates into the inside. The undercuts **121S** may cause instability in the oxidation process for forming the barrier oxide **125'**.

[0080] After all, the surface energy of the undercut **121S** may be increased, making the oxidized thickness thin, and the source channel contact **121''** and the channel layer **110** may be damaged in the subsequent process.

[0081] According to the embodiments, horizontal recesses of a high aspect ratio may be filled with polysilicon containing different dopants without voids.

[0082] According to the embodiments, since polysilicon is formed with a difference in wet etch rates, an etched surface may be protected from being undercut.

[0083] According to the embodiments, the reliability of the vertical semiconductor device may be improved.

[0084] While the embodiments have been described with respect to the specific embodiments, it will be apparent to those skilled in the art that various changes and modifications may be made without departing from the spirit and scope of the description as defined in the following claims.

What is claimed is:

1. A vertical semiconductor device, comprising:

a lower stack including a source channel contact, wherein the source channel contact includes a first conductive layer and a second conductive layer;

an upper stack including a plurality of dielectric layers and a plurality of gate electrodes alternately stacked over the lower stack;

a channel layer penetrating the upper stack and extending in the lower stack, wherein the channel layer contacts with the source channel contact;

a barrier oxide formed over the first and second conductive layers of the source channel contact;

a vertical contact recess formed in the upper stack and extending to the barrier oxide; and

a sealing spacer formed over a side wall of the vertical contact recess and extending to cover the barrier oxide, wherein the barrier oxide includes a non-planar surface in contact with the first and second conductive layers of the source channel contact, and a planar surface in contact with the sealing spacer.

2. The vertical semiconductor device of claim 1,

wherein the source channel contact includes a concave portion and a convex portion, the concave portion and the convex portion are directly contacted with the barrier oxide,

wherein the convex portion of the source channel contact has a peak, and the peak is provided by the second conductive layer; and

wherein the concave portion and the convex portion of the source channel contact are fully covered by the barrier oxide.

3. The vertical semiconductor device of claim 1, wherein the first conductive layer is positioned between the channel layer and the second conductive layer, and the first conductive layer has a shape surrounding the second conductive layer.

4. The vertical semiconductor device of claim 1, wherein the second conductive layer includes a material having a wet etch rate different from the first conductive layer.

5. The vertical semiconductor device of claim 1, wherein the first conductive layer includes polysilicon containing a first dopant, and the second conductive layer includes polysilicon containing a second dopant, where the first dopant and the second dopant are different materials.

6. The vertical semiconductor device of claim 1, wherein the second conductive layer includes carbon-doped polysilicon.

7. The vertical semiconductor device of claim 5, wherein the first conductive layer includes phosphorus-doped polysilicon.

8. The vertical semiconductor device of claim 1, wherein the second conductive layer has a thickness that is thinner than a thickness of the first conductive layer.

9. The vertical semiconductor device of claim 1, wherein the barrier oxide includes silicon oxide.

10. The vertical semiconductor device of claim 1, wherein the second conductive layer includes carbon-doped polysilicon, the first conductive layer includes phosphorus-doped polysilicon, and the barrier oxide includes silicon oxide.

11. The vertical semiconductor device of claim 2, wherein the barrier oxide directly contacts the concave portion and the convex portion.

12. The vertical semiconductor device of claim 1, wherein the second conductive layer includes a first end and a second end,

wherein the first end of the second conductive layer is laterally spaced from the channel layer by the first conductive layer, and

wherein the second end of the second conductive layer is fully covered by the barrier oxide.

13. The vertical semiconductor device of claim 1, wherein the lower stack includes:

a lower source layer; and

an upper source layer over the lower source layer,

wherein the source channel contact is disposed between the lower source layer and the upper source layer.

14. The vertical semiconductor device of claim 1, further comprising:

a vertical source line disposed in the vertical contact recess.

15. The vertical semiconductor device of claim 1, further comprising:

a tunnel dielectric layer on the channel layer;

a charge storage layer on the tunnel dielectric layer; and

a blocking layer on the charge storage layer,

wherein the tunnel dielectric layer, the charge storage layer and the blocking layer are penetrating the upper stack and extending in the lower stack.

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